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Applications of Onboard AI Processing for Autonomous Space Systems

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Abstract

Small satellites in low Earth orbit may need to respond to solar energetic particle events faster than a ground command loop permits, but continuous learned inference is costly on a spacecraft processor. This paper evaluates a two-layer onboard pipeline using seven years of MetOp-C particle counts. An always-on finite-state trigger maintains a rolling robust background and wakes a small temporal convolutional classifier that uses the same z-score stream. The software follows core Flight System (cFS) Limit Checker semantics and a SUCHAI-style command executor. Against the National Oceanic and Atmospheric Administration (NOAA) solar proton event catalog and locally hand-labeled enhancements, the classifier is the stronger detector: at the multi-channel operating point it reaches event-level $F1 = 0.870$ versus 0.741 for the trigger, with the same probability of detection (0.952) and about half the event-level false alarm ratio. The trigger remains useful because it limits compute. A two-sample gate opens on 6.84% of samples, about 6.6 times per day, and reduces classifier compute by a factor of 14.6. One classifier evaluation costs 374 trigger evaluations, yet the gated classifier uses 13.8 ms and about 11 mJ per day. The connected pipeline reveals an interface constraint: most gate openings are shorter than the classifier's four-sample decision rule, reducing end-to-end probability of detection from 0.952 to 0.690. Wake-up and output aggregation must be chosen together: a joint sweep shows that detection follows the combined persistence of the two rules while compute follows only the trigger's, so the same detection performance costs 0.39 or 17.4 classifier calls per day depending on how the persistence is split. Placing the persistence in the always-on layer reaches event-level $F1$ 0.864 at 0.39 calls per day, against 96 for a classifier evaluated at every sample. The catalog and the local labels are best read as two different event conventions, not as one ground truth.

Keywords: onboard autonomy, solar energetic particles, low Earth orbit, temporal convolutional network, CubeSat, edge computing

Acronyms/Abbreviations

cFS	core Flight System
FAR	false alarm ratio
LEO	low Earth orbit
MAD	median absolute deviation
MEPED	Medium Energy Proton and Electron Detector
NOAA	National Oceanic and Atmospheric Administration
OOF	out-of-fold
POD	probability of detection
ROS 2	Robot Operating System 2
SAA	South Atlantic Anomaly
SEM-2	Space Environment Monitor-2
SEP	solar energetic particle
TCN	temporal convolutional network

1. Introduction

1.1 Radiation hazards in LEO and the limits of the ground loop

Small satellites in low Earth orbit (LEO) operate in a damaging and poorly predictable radiation environment. Energetic particles deposit total ionizing dose and induce single-event effects, and a single-event latch-up can destroy a device in seconds [1]; several CubeSat missions have ended early for this reason [2], [3], [4]. Solar energetic particle (SEP) events can reach LEO within tens of minutes of a solar eruption, and the South Atlantic Anomaly (SAA) raises trapped-particle exposure on every transit.

Normal CubeSat operations cannot respond on that time scale. With ultra-high-frequency links near 9.6 kbit/s and only a few short contacts per day, the spacecraft goes unsupervised for most of each day [2], and a protective action needed within seconds cannot wait for downlink, analysis, and uplink. The mission concept considered here has no alternative to autonomous onboard detection.

1.2 Two literatures that do not meet

Machine learning for SEP prediction has advanced quickly, but mainly on the ground and from solar inputs: recent models use flare, coronal mass ejection, and active-region features [5], [6] under a common validation protocol [7]. Those precursors are not observed by a local particle detector. Onboard autonomy has instead focused on optical inference [8], post-hoc telemetry analysis [9], and model compression for CubeSat-class processors [10]. A one-dimensional particle stream screened continuously for the onset of a local enhancement is a different problem and has not been demonstrated in orbit. Our earlier work on this testbed established the Robot Operating System 2 (ROS 2) and PX4 integration by gating an object detector, but addressed a camera-based perception task.

1.3 Gating as a cost argument

When onboard inference must fit a power budget, a common edge-computing pattern is an inexpensive always-on front end that wakes a stronger back end only when needed [11]. The front end need not be the better detector; it must be cheap and prioritize event coverage, and average cost is governed mainly by how often the gate opens. Particle monitoring suits this pattern well: event intervals are sparse, an unnecessary opening costs one forward pass, and a missed opening removes the event from every downstream model. That leaves three questions for the design, which this paper takes in turn: what the gate costs, how often it opens, and what it hands to the classifier.

This paper evaluates a two-layer pipeline built on that principle. A rule-based trigger runs on raw counts and maintains a rolling robust background; when it fires, a small classifier labels the current state from a z-score window drawn from the same background. Both layers are implemented as ROS 2 nodes. Their interfaces follow the core Flight System (cFS) Limit Checker and Stored Command applications [12], with runtime and persistence mapped to a SUCHAI-style command executor and data repository [13].

1.4 Contributions

1. **A matched-condition comparison** of a rule-based trigger and a learned classifier on the same LEO count series, using two reference event sets and two channel configurations.
2. **A characterization of the two reference sets**, which differ in event granularity and onset definition more than in simple coverage.
3. **A gate treated as an operating point rather than a standalone detector**, with persistence selected from its event coverage and duty cycle.
4. **Measured onboard cost on an edge testbed**, together with an end-to-end evaluation that identifies a persistence mismatch between the gate and the classifier's decision rule.

Section 2 describes the data and reference events, Section 3 the onboard pipeline and evaluation protocol, Section 4 the results, and Sections 5 and 6 the interpretation and conclusions. Two appendices list the two event sets in full.

2. Data and Labels

2.1 Instrument and count series

The particle data come from the Medium Energy Proton and Electron Detector (MEPED) in the Space Environment Monitor-2 (SEM-2) package aboard MetOp-C, a Sun-synchronous polar orbiter at about 830 km with a period near 100 min [14], [15]. MEPED has four solid-state telescopes that measure protons and electrons at 0 degrees (zenith) and 90 degrees, and four omnidirectional proton domes [16].

Table 1 lists the nominal energy bands from [14], [15], and [17]. Each species-direction-energy combination is treated as one count series, giving 22 physical channels. The label anchor used throughout, `omni_p6`, is the lowest-threshold dome channel and responds to protons above a nominal 16 MeV.

We work from the raw count product, not the processed flux product that the instrument documentation recommends for scientific analysis [16]. Counts are available onboard; calibrated flux requires instrument-specific tables, energy-band assumptions, and cross-channel corrections carried out on the ground. A trigger validated on flux would never see that input in flight.

Three instrument limits affect interpretation. Directional P6 can be contaminated by relativistic electrons when high-energy solar protons are absent [18]. The dome channels mix solar protons with trapped particles,

including SAA passages [15]. Detector degradation, cross-species contamination, and inter-satellite calibration differences also remain [16]; detector aging motivates the separate NOAA-19 comparison below.

The record spans 1 January 2019 through 31 December 2025, beginning with the first complete calendar year after SEM-2 activation [15]. One-minute counts are averaged to a uniform 15-min cadence, which suppresses isolated noise while retaining onset changes on time scales of tens of minutes to hours. For omni_p6, this reduces 3,546,091 samples to 236,848.

NOAA-19 carries an instrument of the same design but was launched nine years earlier [15]. It is used only for the rule-based transfer check in Section 4.1 because the hand labels exist only for MetOp-C; its proton detectors have accumulated more fluence degradation [19].

Table 1. MEPED channels, look directions, and nominal energy bands [14], [15], [17].

Detector	Species	Look direction	Channels	Nominal energy band
Telescope	Proton	0° (zenith)	P1–P6	30–80, 80–240, 240–800, 800–2500, 2500–6900 keV; P6 >6900 keV
Telescope	Proton	90°	P1–P6	(same as above)
Telescope	Electron	0° (zenith)	E1–E3	>30, >100, >300 keV (nominal); >43, >114, >292 keV (revised)
Telescope	Electron	90°	E1–E3	(same as above)
Dome	Proton	Omnidirectional (4 domes)	P6–P9	>16, >35, >70, >140 MeV; upper limit $\approx$ 250 MeV

2.2 Reference event sets

Detections are scored against two independent event sets. The first is the National Oceanic and Atmospheric Administration (NOAA) solar proton event catalog, defined on the >10 MeV integral proton channel at 10 proton flux units (pfu); it contains 42 events in the study interval. The catalog is an operational standard, but it is measured at geostationary altitude and does not share the anchor channel's 16 MeV threshold.

The second set comes from direct reading of the omni_p6 count envelope. One analyst marked onset, peak, and end for each candidate enhancement, yielding 53 complete events; eight have a peak count below two and are flagged as ambiguous. The full list is given in Table A1. No alternative anchor channel was evaluated, so the labels inherit the visibility limits of omni_p6.

Correspondence depends on the criterion. If a catalog start falls inside a hand-labeled interval extended by the tolerance, 40 of 42 catalog events have a counterpart. Under the one-to-one onset matching used for detector scoring, 13 catalog events remain unmatched. Table B1 lists the catalog entries and the counterpart found for each. The sets disagree less on coverage than on where an event begins and how many events an enhancement contains.

Hand-labeled onsets precede catalog starts by a median 8.5 h and by as much as 22.2 h, because the labels mark the visible rise in local counts while the catalog begins at a geostationary flux threshold. At ± 12 h, seven otherwise clear interval matches would be lost. Six hand-labeled enhancements contain two or three catalog events, while 15 contain none. In short, the catalog is later and finer in event granularity; the hand labels are earlier and broader. Table 2 summarizes the correspondence, and Fig. 1 shows one representative case of each type.

The three marked times define quiet, rising, and decreasing states on the 15-min grid. The full record contains 91.32% quiet, 2.24% rising, and 6.44% decreasing samples, so event intervals occupy 8.68% of the record. Since the trigger emits an onset and not a graded state, rising and decreasing are merged into one event class for classifier training.

Table 2. Reference event sets and their correspondence, 2019–2025.

	Count
NOAA catalog (>10 MeV, 10 pfu)	42
Hand-labeled enhancements (complete onset/peak/end)	53
— of which peak count below two	8
Interval correspondence (catalog begin inside hand interval ± 24 h)	
catalog events with a hand-labeled counterpart	40
catalog events without	2
hand-labeled enhancements containing at least one catalog event	38
hand-labeled enhancements containing none	15
hand-labeled enhancements containing two or more	6
Onset correspondence (one-to-one, ± 24 h; used for detector scoring)	

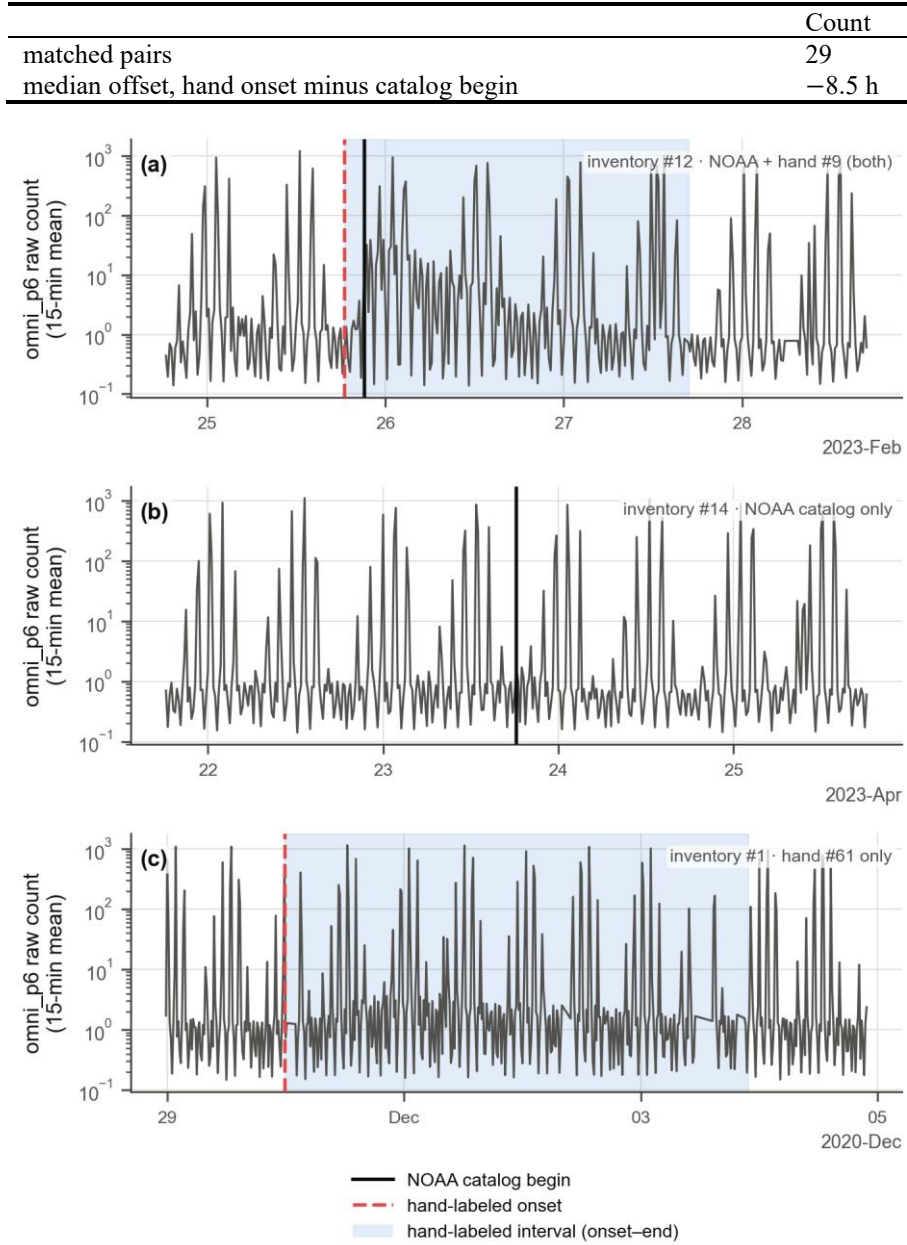


Fig. 1. Representative cases from the two reference event sets.

2.3 Supervised dataset construction

The classifier uses fixed-length windows of the z-score series defined in Section 3.1. Each window contains 14 consecutive 15-min samples (210 min, about two orbital periods) and is labeled by its final sample, making the task a nowcast at the window end.

Training windows are retained within ± 10 days of a labeled event to prevent distant quiet periods from dominating the loss. Overlapping padded intervals are merged before splitting because adjacent stride-one windows share 13 of 14 samples; otherwise nearly identical windows could cross the train-validation boundary. The 53 events form 22 episodes. Of 236,835 candidate windows, 96 lack complete z-score coverage and 157,640 fall outside the crop, leaving 79,099 windows that are 74.00% quiet and 26.00% event. Every channel configuration uses the same crop.

Model selection and evaluation use grouped five-fold cross-validation with episode as the grouping unit. All classifier metrics are out-of-fold (OOF): each window is scored by a model that did not see its episode during training. The trigger requires no training and is evaluated over the full record.

3. Two-Layer Onboard Architecture

3.1 Layer 1: a flux-free finite-state trigger

The first layer runs at every cadence and tests each count against a slowly varying background. Because LEO particle counts drift with solar activity and vary with SAA transits and high-latitude precipitation, a fixed threshold is unsuitable; the trigger estimates its reference from a trailing window.

For a channel count series $c(t)$, the background is summarized by the median m and robust scale

$$\sigma = 1.4826 \cdot \text{median}(|x - \text{median}(x)|) \quad (1)$$

over the samples of the preceding W days, and the detection threshold is

$$T(t) = \max(m + k \sigma, f) \quad (2)$$

Here x denotes samples in the preceding W days, k is the threshold multiplier, and f is the minimum count floor.

The median absolute deviation (MAD) is used because a single spike can destabilize a standard-deviation threshold. The operating point (W , k , f) is selected in Section 4.1. The background is recomputed once per day and held between updates: a seven-day window contains 672 samples, so one new sample replaces only 0.15% of it. This keeps the median and MAD calculations off the recurring per-sample path.

A candidate event is a run of consecutive samples satisfying $c(t) \geq T(t)$. The offline detector accepts a completed run lasting at least one hour; the onboard form counts consecutive exceedances causally. Section 4.6 compares the two forms.

The z-score consumed by the second layer is formed from the same two statistics,

$$z(t) = \text{clip}((c(t) - m) / \max(\sigma, \epsilon), -Z, +Z) \quad (3)$$

with $\epsilon = 1$ count and $Z = 10$. The floor prevents division by a near-zero σ during quiet intervals, and clipping limits the effect of a single extreme sample.

3.2 Layer 2: a small temporal convolutional classifier

The second layer labels the current state from a z-score window. We use a temporal convolutional network (TCN) [20] because causality is built into the convolutions, per-window cost is fixed, and no recurrent state must be restored when the gate opens. Networks of similar scale have been demonstrated on embedded targets [21].

The network has three residual blocks with 16 filters, kernel size 3, dropout 0.2, and dilations 1, 2, and 4. Two causal convolutions per block give a 29-sample receptive field, covering the 14-sample input. The single-channel binary model has about 4.0×10^3 parameters, 16 KiB of 32-bit weights, and 74,752 multiply-accumulate operations per window. Additional channels enlarge only the first convolution.

The classifier receives no measurement unavailable to the trigger. The background statistics of Eq. (1) and the resampled counts are already held by Layer 1, so the marginal work is window assembly and the forward pass.

3.3 Gating and the two-stage alert

The layers do not run in parallel: Layer 1 is always on, and Layer 2 wakes only when the trigger raises its state. An unnecessary opening costs one forward pass; a missed opening hides the event from the classifier. That asymmetry is what makes probability of detection the criterion for selecting the gate, with persistence as the free parameter (Section 4.4).

Two states are emitted at different persistence levels:

- PRE_ALERT is raised after a short run of exceedances. It wakes the classifier, begins setpoint pre-streaming, and would support an onboard beacon. It is reversible: one sample below threshold returns the state to nominal.
- ALERT is raised only after the longer confirmation rule. It is latched and maps to a vehicle mode transition.

The two decisions differ in cost: waking a classifier is cheap and reversible, while commanding a mode change is not. They also use different persistence. The gate opens after two consecutive exceedances, but classifier probabilities are converted to detections with a four-sample rule. An opening shorter than four samples cannot produce a classifier detection at all. Section 4.5 measures this interaction.

3.4 Realisation as flight software

The **trigger semantics** follow the cFS Limit Checker and Stored Command applications [12]. Threshold results are combined before persistence, actionpoints can be active, passive, or disabled, and the passive state preserves evaluation and publication while suppressing the latched action. Because the passive state still evaluates and publishes, the cost measurements cover the full decision and message path.

The **runtime and persistence model** follows SUCHAI [13]. Node actions are registered as commands, and the seven-day background and classifier input buffer are checkpointed to nonvolatile storage with an integrity check; a failed check starts from a cold background instead of a corrupted threshold.

A fourth node maps a latched alert to a vehicle mode transition but runs with commands suppressed. This paper measures the response path; it does not demonstrate closed-loop control.

Fig. 2 shows the four nodes and what passes between them. WP resamples the 1-min stream, maintains the rolling median and MAD, and tests each count against the threshold of Eq. (2); the same background estimate forms the z-score the classifier consumes, so the two layers share one estimate instead of each computing its own. WP and AP evaluate every tick, AI runs only while the gate is open, and SC acts only on a latched alert. The flight target is a Zynq-7030 with a dual-core ARM Cortex-A9 at up to 800 MHz and no GPU [22]. Cost measurements are single-threaded, and timing instrumentation is kept outside the detection logic so it cannot affect a decision.

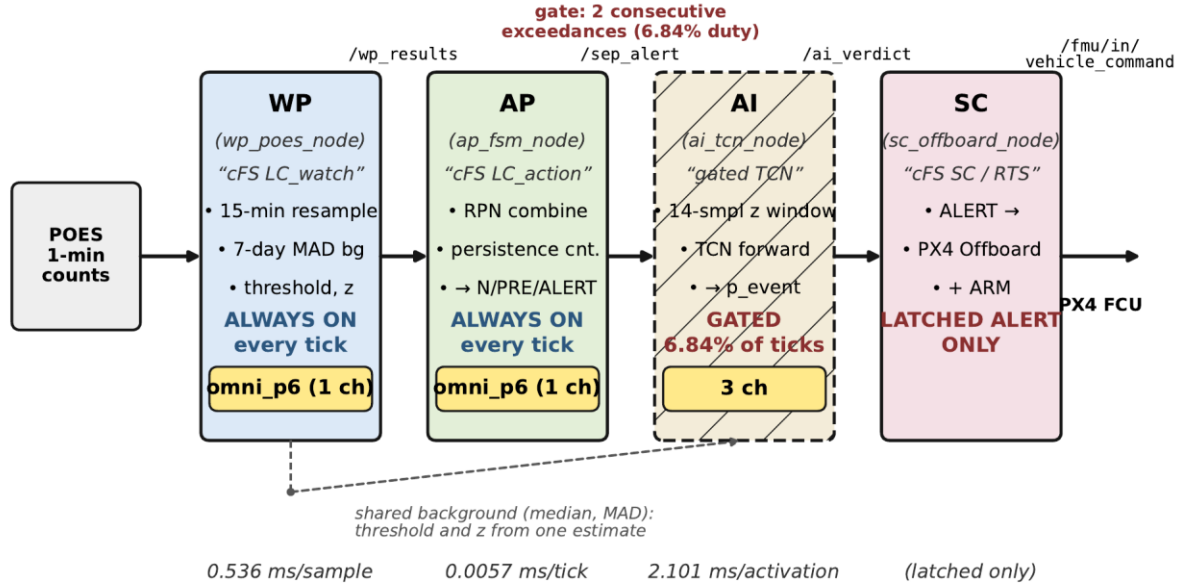


Fig. 2. The onboard pipeline from raw counts to a vehicle command. Costs below each block are the bare medians of Table 7.

3.5 Evaluation protocol

Detections are matched to reference events within ± 24 h, set by the observed onset difference: common events differ by a median 8.5 h and by up to 22.2 h. Matching is one-to-one and greedy, with candidate pairs ordered by absolute offset. We report probability of detection (POD), the fraction of reference events matched; false alarm ratio (FAR), the fraction of detections unmatched; and the median signed offset. Detector scoring matches onset to onset, whereas Section 2.2 tests whether a catalog start falls inside a hand-labeled interval.

Repeated threshold crossings from one physical event are clustered on the same 24-h scale before scoring, so event-level FAR is computed over clusters rather than raw detections. The same matching and clustering are used for both layers and both reference sets.

Classifier probabilities are converted to detections by the trigger's segment rule with a fixed probability threshold. The threshold maximizing event-level F1 is reported.

4. Results

4.1 Trigger configuration

We swept trailing-window length W from 1 to 30 days and threshold multiplier k from 0 to 20 on omni_p6, giving 56 configurations scored against the NOAA catalog (Fig. 3). Curves for $W = 5$ -30 days are similar, so $W = 7$ days is chosen as the shortest window on the performance plateau. Raising k from 5 to 7 reduces event-level FAR from 0.200 to 0.114 with a 0.024 loss in POD; $k = 10$ lowers POD to 0.690 without a further FAR reduction. We therefore use $W = 7$ days, $k = 7$, and $f = 0.1$ count. The floor is inactive on omni_p6.

At this operating point the trigger produces 224 onsets, matches 31 of 42 catalog events, and yields POD 0.738, event-level FAR 0.114, and precision 0.886. Applied unchanged to NOAA-19, it gives POD 0.619 and FAR 0.469, consistent with the older detector's accumulated degradation [19].

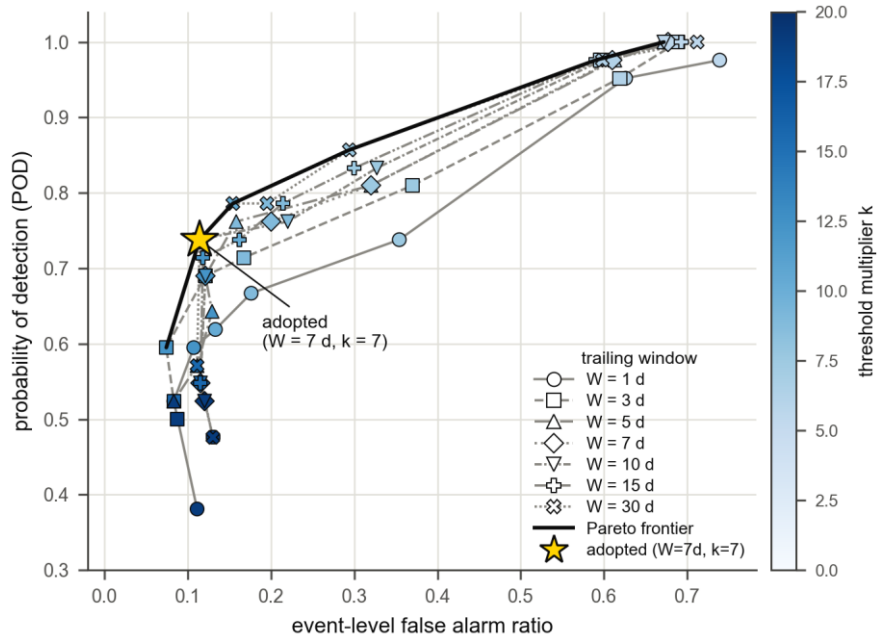


Fig. 3. Trigger operating surface over window length and threshold multiplier.

4.2 Detection performance of the two layers

Table 3 compares the trigger and classifier under the same record, matching, and clustering rules.

Table 3. Detection performance under matched scoring. Trigger rows use the operating point of Section 4.1; TCN rows use the F1-maximizing probability threshold.

Layer	Channels	Reference	threshold	POD	event FAR	Precision	F1
Trigger	single	NOAA	—	0.738	0.114	0.886	0.805
TCN	single	NOAA	0.98	0.976	0.255	0.745	0.845
Trigger	multi	NOAA	—	0.952	0.394	0.606	0.741
TCN	multi	NOAA	0.99	0.952	0.2	0.8	0.870
TCN	multi	NOAA	0.999	0.69	0.094	0.906	0.784
Trigger	single	hand	—	0.453	0.226	0.774	0.571
TCN	single	hand	0.97	0.83	0.267	0.733	0.779
Trigger	multi	hand	—	0.623	0.476	0.524	0.569
TCN	multi	hand	0.85	0.868	0.452	0.548	0.672

The classifier has higher F1 in every paired comparison. In the multi-channel NOAA comparison, both layers reach POD 0.952, but the classifier reduces event-level FAR from 0.394 to 0.200 and reaches F1 0.870 versus 0.741. Additional channels therefore help the classifier but harm the trigger, whose logical union also combines channel-specific excursions.

Scores are lower against the hand labels because the two references use different onset and event granularity. The single-channel classifier also detects slightly earlier relative to catalog onset than the trigger (+0.75 h versus +0.92 h). Trigger parameters were selected against the catalog while classifier training used hand labels, which makes the cross-evaluated cells the least favorable comparison for either method.

4.3 Channel configuration

Table 4. Out-of-fold classifier performance by input-channel set.

Channels	n	Macro-F1	F1 (event)	Precision (event)	Recall (event)
omni_p6	1	0.848	0.778	0.756	0.802
omni_p6, pro_tel90_p5	2	0.872	0.809	0.832	0.788
omni_p6, omni_p7	2	0.873	0.813	0.803	0.826
omni_p6, omni_p7, pro_tel0_p5, pro_tel90_p5	4	0.874	0.815	0.808	0.827
omni_p6, omni_p7, pro_tel0_p5	3	0.876	0.818	0.804	0.836
omni_p6, pro_tel0_p5, pro_tel90_p5	3	0.879	0.82	0.833	0.807

Table 4 gives out-of-fold performance for each input-channel set. All five multi-channel inputs improve macro-F1 by 0.024-0.031 and event-class F1 by 0.031-0.042 over omni_p6 alone, while differences among the multi-channel sets remain within fold-to-fold variation. The four-channel set offers no further gain.

Subsequent classifier results use omni_p6 with the 0° and 90° P5 proton telescopes. The added channels share one energy band and provide complementary look directions for trapped and precipitating populations.

4.4 Gate selection

Gate persistence N was swept from one to four consecutive exceedances, and duty cycle was measured as the fraction of samples at which the classifier would run (Table 5).

Table 5. Gate persistence, duty cycle, and event coverage. Calls/day is the duty cycle scaled to the 96 samples of a complete day; the recorded series contains gaps, so the calendar-day rate at N = 2 is 6.34.

Gate	Duty cycle	× base rate	Calls/day	POD (NOAA)	POD (hand)
N = 1	18.08%	2.08	17.4	1	1
N = 2	6.84%	0.79	6.6	1	1
N = 3	0.41%	0.05	0.39	0.833	0.566
N = 4 (confirm)	0.19%	0.02	0.18	0.738	0.434

The transition is abrupt. N = 1 and N = 2 retain every event in both reference sets, but N = 2 reduces duty cycle from 18.08% to 6.84%, or 6.6 calls per day. At N = 3, POD falls to 0.833 against the catalog and 0.566 against the hand labels. We adopt N = 2 for the measurements that follow. The choice is made on gate coverage and duty cycle alone, and Section 4.5 shows that it is not the joint optimum once the classifier's own decision rule is taken into account. Its duty cycle can sit below the 8.68% event sample fraction because event-level POD requires only one matched opening, not coverage of the full event interval.

Unit POD can also be obtained by setting $k = 0$, but event-level FAR rises to 0.677 versus 0.208 for the N = 2 gate. The selected persistence was tuned on this record, so complete coverage is not an estimate for unseen events.

4.5 End-to-end performance of the gated pipeline

Table 6 evaluates the connected pipeline on 236,739 common samples: classifier alone (A), classifier only while the gate is open (B), confirmed trigger with a classifier veto (C), and confirmed trigger alone (D). For B, break treats a closed gate as ending a classifier segment, as in deployment; skip removes closed samples and is an idealized upper bound. In every configuration the gate runs on omni_p6 alone while the classifier receives all three channels. Keeping the extra channels out of the trigger follows from Section 4.2, where they raise classifier F1 but cost the rule precision.

Table 6. End-to-end detection performance against the NOAA catalog.

Configuration	threshold	n det	POD	event FAR	Precision	F1
A. Classifier alone	0.99	217	0.952	0.2	0.8	0.87
D. Confirmation alone	—	216	0.738	0.061	0.939	0.827
C. Confirmation with classifier veto	0.85	212	0.738	0.061	0.939	0.827
B. Gated classifier, skip	0.98	110	0.952	0.298	0.702	0.808
B. Gated classifier, break	0.85	187	0.69	0.065	0.935	0.795

With break semantics, gating lowers POD from 0.952 to 0.690 even though the gate opens for every event. The cause is duration: 15,053 of 15,240 openings (98.8%) are shorter than four samples and cannot satisfy the classifier's segment rule (Fig. 4). Skip preserves POD but raises FAR to 0.298 by making separated openings adjacent, so it is not an implementable configuration.

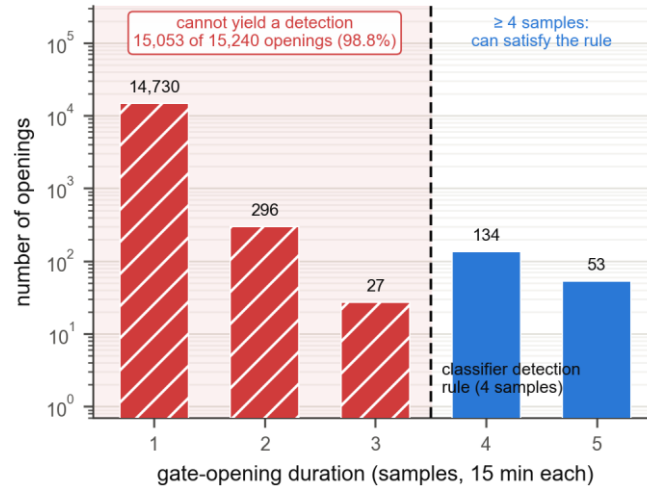


Fig. 4. Gate-opening durations at the selected operating point.

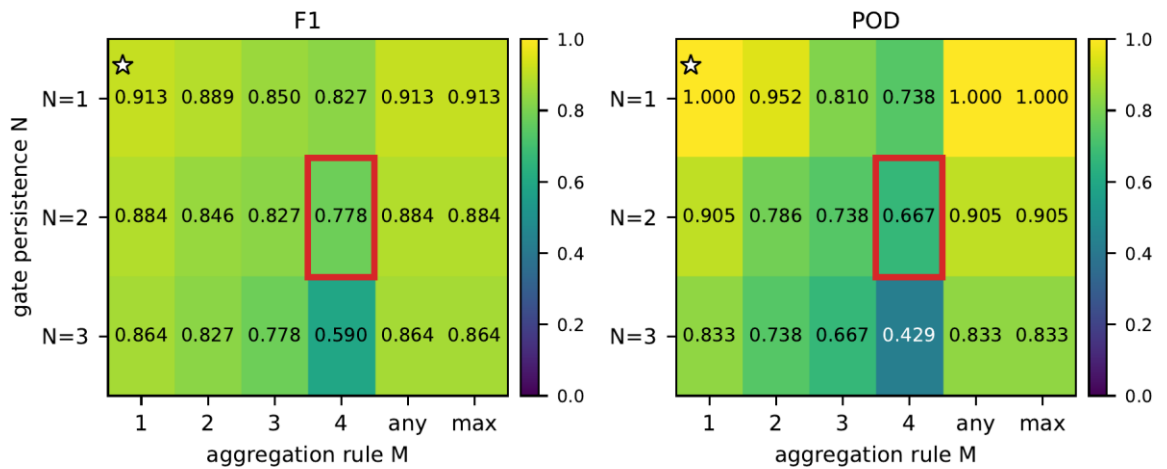
Break lowers event-level FAR to 0.065 because a detection must survive both filters, while skip raises FAR by compressing time. That the two move in opposite directions is what identifies aggregation, and not the loss of gated samples, as the cause of the reduced sensitivity.

The classifier provides no useful veto at trigger confirmation: configurations C and D are identical at event level. Against the hand labels, skip reaches F1 0.702 versus 0.682 for the classifier alone, while the confirmation-based configurations reach 0.568.

Under break, the gate removes 33 matched and 47 unmatched detections, which is not selective. Under skip, it removes 2 matched and 10 unmatched detections. The interface rule, not gate coverage, determines the trade-off.

Section 5.2 argues that the gate and the classifier's aggregation rule have to be chosen together. Fig. 5 sweeps both, with gate persistence N running from one to three exceedances and the aggregation rule M from one to four consecutive positive samples, each cell scored at its own F1-maximizing probability threshold. Two features stand out. Detection follows the total persistence $N + M$ almost exactly: the three cells with $N + M = 5$ return the same POD 0.738 and event-level FAR 0.061, and the two cells with $N + M = 6$ agree in the same way. Cost, in contrast, follows N alone. The same detection performance is therefore available at 17.4, 6.6, or 0.39 classifier calls per day depending only on how the persistence is divided, a range of a factor of 44. The configuration used above, $N = 2$ with the four-sample rule, sits at the expensive end of a high-persistence diagonal. Moving that persistence into the trigger instead, at $N = 3$ with a single-sample rule, raises POD from 0.667 to 0.833 and F1 from 0.778 to 0.864 and cuts the call rate by a factor of 17, at a cost of 0.036 in event-level FAR.

Two limits apply to the grid. It breaks a classifier segment wherever the probability series itself is missing, which is stricter than the alignment behind Table 6, so its $N = 2$, $M = 4$ cell reads POD 0.667 where Table 6 reads 0.690. The per-cell thresholds are also selected on the full record, so the grid supports comparisons between operating points rather than an out-of-sample estimate of any one of them.



red box = current op. point ($N=2$, $M=4$) ★ = optimal (max-F1 cell, $N=1$, $M=1$) each cell at its own F1-maximizing threshold, reference=NOAA

Fig. 5. Event-level F1 and POD over gate persistence N and classifier aggregation M . The red box marks the operating point of Table 6 and the star the highest-F1 cell.

4.6 Onboard cost

Cost was measured on an NVIDIA Jetson Orin Nano at locked clocks with one CPU core and one tensor thread. All six cores were held at 1.51 GHz in the 15 W power mode for every measurement reported here. Bare tests invoke operations directly; in-node tests run the same operations in the four-node ROS 2 pipeline under accelerated replay. Energy was measured separately in single-core bursts because replay power is not representative of flight.

Table 7. Bare per-operation cost; values are medians with 95th percentiles in parentheses.

Operation	Invoked	Repetitions	Cost (ms)
z-score buffer append	every tick	10 ⁵	0.00093 (0.00096)
Trigger evaluation (logic, persistence, state)	every tick	10 ⁵	0.0057 (0.0077)
Alert serialization on activation	on activation	10 ⁴	0.0073 (0.0077)
z-score computation	every tick	10 ⁵	0.0314 (0.0320)
Window assembly for the classifier	on activation	10 ⁴	0.0496 (0.0541)
Background update	once per day	10 ⁴	0.173 (0.191)
Resampling to the 15-minute cadence	every raw sample	10 ⁴	0.536 (0.592)
Classifier forward pass	on activation	10 ³	2.132 (2.319)

One classifier forward pass takes 2.132 ms, 374 times the 0.0057 ms trigger evaluation. Adding window assembly and serialization brings a single-channel activation to 2.189 ms, against 0.038 ms for the per-tick decision path, a factor of 58.

Input width has no measurable timing effect: one-, three-, and five-channel passes take 2.069, 2.044, and 2.052 ms. Evaluating the five-fold ensemble multiplies cost by 5.1.

At $N = 2$, the classifier runs 6.6 times per day and costs 13.8 ms/day, versus 201.6 ms/day if run at every sample (Table 8), a 14.6-fold reduction. The always-on layer costs 776.3 ms/day, of which 772.5 ms is resampling; the gated classifier is 1.8% of that total. In-node execution raises the complete always-on path to 1.36 s/day.

Table 8. Daily compute budget under bare and in-node accounting.

	bare (ms/day)	in-node (ms/day)
Always-on layer, total	776.3	1355.4
— resampling the 1-minute stream	772.5	1339.9
— z-score and rule evaluation (bare column also includes the buffer append)	3.6	14.9
— background update	0.17	0.6
Classifier, gated at the operating point of Section 4.4	13.8	24.2
Classifier, hypothetically always on	201.6	353.0

Table 9. Bare and in-node operation costs. The first four rows are single-channel; the activation row is three-channel, the configuration used in Section 4.5.

Operation	bare (ms)	in-node (ms)	ratio
Resampling to the 15-minute cadence	0.536	0.931	1.7
z-score computation	0.0314	0.129	4.1
Trigger evaluation	0.0057	0.026	4.6
Background update	0.173	0.599	3.5
Classifier activation (assembly, forward pass, serialization), three channels	2.101	3.677	1.8

ROS 2 inflates operation cost by factors of 1.7–4.6 (Table 9), with the largest relative increase on the cheapest operations. A complete three-channel activation takes 3.677 ms in the pipeline against 2.101 ms for the same composition measured bare.

A classifier evaluation consumes 1.684 mJ. The trigger estimate, at most 0.036 mJ, is an upper bound because the operation is shorter than the surrounding harness. At the selected gate rate, classifier energy is about 11.1 mJ/day and trigger energy at most 3.5 mJ/day, that is 3.1 and 1.0 μ Wh per day, or continuous draws of 0.13 and 0.04 μ W. Keeping the two nodes resident costs 44.1 mW above the message bus, so the energy of the detection arithmetic is about one part in 260,000 of the energy needed to host the software that performs it. Per-operation energy is reported because the classifier is not continuously active; an average power figure alone would hide how the cost scales with the gate rate.

Direct timing and process CPU time agree within 2% for inference. Across repeated scenarios, per-operation medians vary by at most 4.8%, and by 1.4% for the classifier activation over four repetitions of the same window; 1800× and 7200× replay agree within 5.5–10.7%.

The in-node tests use three 17-day windows, each running from 10 days before an onset to 7 days after it, spanning a dense event cluster, the strongest event, and the weakest completed event. These windows also give an independent check on the duty cycle that the daily budget assumes. Recomputed offline, the gate is open on 6.3%, 6.3%, and 8.0% of the samples they contain, against 6.84% over the full record, so the three short spans are representative of the rate used in Table 8 rather than unusually quiet or unusually active.

Table 10. In-node resource use during 7200× replay.

Scenario	Nodes	Runs	Activ.	Power (mW)	ΔP idle (mW)	ΔP bus (mW)	CPU (% core)	Δ RAM (MB)
Idle, no nodes	0	3	—	4091.6	—	−38.7	1.7	—
Idle with the message bus	4	3	—	4130.3	38.7	—	6.1	322
Trigger only, 1 channel	2	6	0	4174.4	82.8	44.1	11.4	452
Gated pipeline, 1 channel	3	6	558	4205.2	113.6	74.9	14.8	676
Gated pipeline, 3 channels	3	3	436	4226.4	134.8	96.1	16.7	872
Gated pipeline with command node, 1 channel	4	3	288	4232.5	140.9	102.2	19.4	735

Table 10 reports resource use for each scenario. Loading the four nodes before replay adds 38.7 mW and 322 MB. The three-channel pipeline reaches 872 MB, close to the 1 GB flight configuration, showing that Python and ROS 2 dominate memory. The command-node power row is an upper bound from a revision that repeated commands each control cycle.

The same replay windows also support a single-channel gate. Running causally on the Jetson, it opens 100, 98, and 90 times in the three windows, while the three-channel logical union opens 134, 165, and 137 times, between 1.3 and 1.7 times as often. The extra channels therefore cost little in wake-ups; the penalty for placing them in the trigger is the event-level false alarm ratio of Table 3, which rises from 0.114 to 0.394.

The measured 2.132 ms inference time is dominated by framework dispatch, not by the 74,752 multiply-accumulate operations. It characterizes this Python/ROS 2 implementation, not the network in isolation, and should not be extrapolated to the flight processor by clock ratio alone.

The causal confirmation admits 216 of 224 offline onsets. All eight rejected detections are unmatched, so POD remains 0.738 while event-level FAR falls from 0.114 to 0.061.

5. Discussion

5.1 The layers are not competing detectors

It would be a mistake to read Table 3 as a contest. The TCN is the more accurate detector and it exploits relationships across channels; the trigger earns its place by being cheap enough to run at every sample. What we ask of it is broad event coverage at a low duty cycle, not the best standalone F1. The case for the two-layer design rests on measured cost, not on any claim that the rule detects better.

5.2 The gate is an operating point, not a detector

Because the classifier only ever sees open-gate samples, the trigger's POD sets a ceiling on the pipeline as a whole. Shortening persistence buys that coverage more cheaply than lowering the threshold does. The end-to-end test then exposes a second problem: persistence is applied twice over, once when two trigger samples wake the classifier and again when four classifier samples are needed to declare a detection. Either the classifier should aggregate over whatever opening it is given, or the two rules should be chosen together. Fig. 5 shows what that choice costs and buys. Detection tracks the sum of the two persistences while compute tracks only the first, so persistence is cheapest when it is spent in the layer that runs anyway. At $N = 3$ with a single-sample rule the pipeline reaches F1 0.864, close to what the classifier achieves when run at every sample, using roughly one inference for every 240 that the always-on configuration would need.

5.3 What the learned layer adds that the rule cannot

Two things the learned layer offers have no equivalent in the rule. Extra channels improve the TCN while they degrade the trigger, whose logical union simply accumulates each channel's excursions. And where the

trigger can only report an onset, the classifier reports a state, which is what would allow a return to nominal mode once the event decays.

5.4 Ground truth is not settled for this orbit

The two reference sets are not interchangeable ground truth. The geostationary catalog is later and finer in event granularity; the local LEO labels are earlier and broader. Change the correspondence criterion and the apparent disagreement drops from 13 catalog events to 2, which shows that a detector tuned to the catalog is not necessarily tuned to the local radiation environment.

5.5 Limitations

The pipeline was evaluated by ground replay, not on the flight processor and not in a radiation environment. The causal implementation also departs from the offline reference in completed-run logic and stale-data handling; a flight version would need to bound background age after data gaps. Ground-validated event selectors have failed on raw flight data before [23].

All operating points were selected on the full record. Classifier predictions are out-of-fold, but the probability threshold is not. Training draws windows from near labeled events while evaluation covers the whole record, and the labels come from one analyst working on one anchor channel. We did not evaluate recurrent or plain convolutional baselines.

Timing and memory are implementation-bound: framework and message-bus overhead dominate a network this small, and fixed-point flight software would behave differently from the Python/ROS 2 stack. Scenario power was measured at 7200× replay and is not flight power, and the trigger energy is an upper bound. Gate persistence and classifier aggregation were also selected separately, so the connected result is not a jointly optimized operating point.

6. Conclusions

This study evaluated a two-layer onboard detector on seven years of MetOp-C particle counts. The TCN reached event-level F1 0.870 versus 0.741 for the rule-based trigger in the multi-channel NOAA comparison. The $N = 2$ gate reduced classifier duty cycle to 6.84% and compute by a factor of 14.6, making the stronger detector affordable. The connected test exposed a separate constraint: most gate openings are shorter than the classifier's four-sample rule, reducing POD from 0.952 to 0.690 even though the gate opens for every event. Wake-up and output aggregation have to be designed together. A joint sweep shows what is at stake: detection follows the combined persistence of the two rules while compute follows only the trigger's, so the same detection performance can cost 0.39 or 17.4 classifier calls per day depending on how the persistence is divided. For deployment we would therefore place the persistence in the trigger, at $N = 3$ with a single-sample classifier rule, which reaches event-level F1 0.864 at 0.39 classifier calls per day; the cost figures reported here were measured at $N = 2$, and the recommended configuration remains to be characterised on the flight testbed. The catalog and the local labels also represent different onset and event conventions.

6.1 Future work

A separate forecasting model that adds solar-context inputs is under development. End-to-end drone tests in PX4 Offboard mode are also under way, connecting the detected alert to the command path. Beyond that we intend to select gate persistence and classifier aggregation jointly, test transfer to an aged detector, and measure fixed-point execution on the flight target.

Appendix A: Hand-labeled event list

Table A1. Complete hand-labeled count enhancements on omni_p6 (N = 53), ordered by onset. Entries marked ambiguous have a peak count below two.

event id	onset (UTC)	peak (UTC)	end (UTC)	peak count	ambiguous
61	2020-11-29 23:45	2020-12-01 01:00	2020-12-03 21:45	1011.5	
60	2021-05-28 17:15	2021-05-29 01:00	2021-05-30 05:00	1039.53	
1	2021-10-28 16:00	2021-10-28 22:15	2021-11-04 16:30	93.79	
2	2022-01-20 06:45	2022-01-20 12:30	2022-01-21 08:15	1152.03	
4	2022-03-28 10:30	2022-03-28 16:00	2022-03-29 15:45	15.15	
5	2022-03-30 22:45	2022-03-31 08:30	2022-04-01 04:00	4.08	
6	2022-04-02 07:30	2022-04-02 12:45	2022-04-03 17:00	641.54	
59	2022-07-09 06:15	2022-07-09 14:00	2022-07-10 07:15	217.82	
7	2022-08-27 07:00	2022-08-27 13:30	2022-08-28 05:00	722.06	
8	2023-02-24 20:00	2023-02-25 01:15	2023-02-25 08:30	936.79	
9	2023-02-25 18:30	2023-02-26 01:00	2023-02-27 16:45	946.39	
10	2023-03-12 17:15	2023-03-13 12:00	2023-03-16 17:00	949.0	
12	2023-05-07 19:00	2023-05-09 01:15	2023-05-09 19:00	579.67	
13	2023-05-09 16:45	2023-05-10 13:45	2023-05-12 18:45	762.83	
58	2023-07-10 06:00	2023-07-10 21:15	2023-07-11 15:00	1.0	yes
14	2023-07-17 18:30	2023-07-18 02:00	2023-07-21 08:00	896.97	
56	2023-07-24 16:45	2023-07-25 12:00	2023-07-28 15:30	1.57	yes
15	2023-07-28 12:15	2023-07-29 11:00	2023-08-02 05:30	417.24	
16	2023-08-05 02:45	2023-08-05 12:00	2023-08-07 21:15	844.89	
17	2023-08-07 17:00	2023-08-08 21:45	2023-08-11 17:45	34.31	
18	2023-08-31 18:30	2023-09-01 01:30	2023-09-03 06:30	1014.59	
19	2023-12-14 16:00	2023-12-15 18:45	2023-12-17 05:45	0.76	yes
20	2024-01-01 17:00	2024-01-04 01:45	2024-01-10 17:15	934.85	
55	2024-01-22 06:15	2024-01-22 15:15	2024-01-23 06:00	0.73	yes
21	2024-01-28 18:15	2024-01-29 12:30	2024-02-01 06:15	885.38	
22	2024-02-09 05:30	2024-02-10 02:15	2024-02-12 07:30	799.63	
24	2024-02-12 05:00	2024-02-13 01:15	2024-02-20 09:15	976.83	
25	2024-03-15 04:15	2024-03-16 01:45	2024-03-19 08:30	549.51	
26	2024-03-22 15:15	2024-03-24 14:00	2024-03-31 01:30	188.87	
27	2024-05-09 15:45	2024-05-11 12:00	2024-05-13 11:00	630.66	
28	2024-05-13 07:15	2024-05-14 01:30	2024-05-19 18:30	915.83	
50	2024-05-20 02:30	2024-05-21 00:45	2024-05-25 02:00	819.32	
29	2024-06-07 16:15	2024-06-08 08:30	2024-06-11 22:45	479.55	
49	2024-06-11 17:00	2024-06-12 12:45	2024-06-15 19:45	1123.44	
30	2024-07-22 16:45	2024-07-23 12:00	2024-07-26 18:00	640.22	
31	2024-08-05 06:15	2024-08-05 21:30	2024-08-07 06:15	24.4	
32	2024-09-01 05:15	2024-09-03 12:45	2024-09-06 16:15	3.17	
33	2024-09-08 18:30	2024-09-10 00:45	2024-09-11 05:30	1.29	yes
36	2024-09-14 08:15	2024-09-17 12:30	2024-09-21 05:15	529.87	
54	2024-09-23 17:45	2024-09-24 12:15	2024-09-26 18:00	0.98	yes
37	2024-10-07 16:45	2024-10-10 01:45	2024-10-12 16:30	707.91	
38	2024-10-24 04:30	2024-10-28 02:15	2024-11-05 13:15	683.51	
39	2024-11-21 16:15	2024-11-22 02:00	2024-11-25 08:30	827.19	
40	2024-12-20 16:00	2024-12-22 02:00	2024-12-28 18:45	1.64	yes
41	2025-01-04 15:45	2025-01-05 01:45	2025-01-07 00:30	668.24	
42	2025-02-24 15:00	2025-02-25 01:00	2025-02-27 17:30	655.98	
43	2025-03-30 04:30	2025-04-01 02:30	2025-04-04 06:15	15.97	
48	2025-05-13 06:30	2025-05-13 23:00	2025-05-14 18:30	156.76	
44	2025-05-31 04:45	2025-06-01 09:30	2025-06-08 11:30	57.67	
45	2025-08-21 19:15	2025-08-26 08:30	2025-09-02 20:00	34.28	
52	2025-10-21 16:00	2025-10-22 11:00	2025-10-26 23:30	0.87	yes
53	2025-11-04 19:00	2025-11-05 13:15	2025-11-06 18:45	975.2	
46	2025-11-10 05:00	2025-11-12 01:15	2025-11-17 06:00	761.21	

Appendix B: Catalog correspondence

Table B1. NOAA solar proton event catalog entries (N = 42) and their correspondence with Table A1 under the matching of Section 3.5. Times are as published by SWPC; the analysis uses only the begin time.

#	begin (UTC)	max (UTC)	max (pfu)	matched hand label	onset difference (h)
1	2021-05-29 03:00	2021-05-29 03:20	15.0	60	-9.75
2	2021-10-28 16:35	2021-10-29 02:50	29.0	1	-0.58
3	2021-10-30 21:00	2021-10-30 21:05	11.0	-	-
4	2022-01-20 08:00	2022-01-20 10:15	22.0	2	-1.25
5	2022-03-28 13:25	2022-03-28 14:50	19.0	4	-2.92
6	2022-03-31 06:20	2022-03-31 06:30	11.0	5	-7.58
7	2022-04-02 14:30	2022-04-02 16:00	32.0	6	-7.00
8	2022-08-27 11:55	2022-08-27 12:20	27.0	7	-4.92
9	2023-02-25 21:10	2023-02-26 04:40	58.0	9	-2.67
10	2023-03-13 07:35	2023-03-15 04:25	22.0	10	-14.33
11	2023-04-23 18:15	2023-04-23 18:20	26.0	-	-
12	2023-05-08 12:40	2023-05-09 01:50	38.0	12	-17.67

#	begin (UTC)	max (UTC)	max (pfu)	matched hand label	onset difference (h)
13	2023-05-09 23:35	2023-05-10 12:50	83.0	13	-6.83
14	2023-07-16 06:35	2023-07-16 07:35	18.0	-	-
15	2023-07-18 01:15	2023-07-18 06:15	620.0	14	-6.75
16	2023-07-29 00:20	2023-07-29 09:20	154.0	15	-12.08
17	2023-08-05 11:15	2023-08-05 17:45	18.0	16	-8.50
18	2023-08-08 01:15	2023-08-09 00:25	47.0	17	-8.25
19	2023-09-01 04:30	2023-09-01 06:10	25.0	18	-10.00
20	2023-12-15 23:45	2023-12-16 00:15	13.0	-	-
21	2024-01-03 20:05	2024-01-04 08:35	20.0	-	-
22	2024-01-29 06:15	2024-01-29 18:05	137.0	21	-12.00
23	2024-02-09 15:30	2024-02-09 23:55	187.0	22	-10.00
24	2024-02-12 08:05	2024-02-13 16:05	118.0	24	-3.08
25	2024-03-23 08:15	2024-03-23 18:20	956.0	26	-17.00
26	2024-05-10 13:35	2024-05-10 17:45	208.0	27	-21.83
27	2024-05-11 02:10	2024-05-11 09:10	116.0	-	-
28	2024-06-08 02:55	2024-06-08 08:00	1030.0	29	-10.67
29	2024-07-23 03:00	2024-07-23 10:40	24.0	30	-10.25
30	2024-09-09 16:40	2024-09-09 22:50	34.0	33	-22.17
31	2024-09-17 07:35	2024-09-17 10:50	33.0	-	-
32	2024-10-09 05:05	2024-10-10 15:15	1810.0	-	-
33	2024-10-26 19:10	2024-10-28 03:50	364.0	-	-
34	2024-11-21 19:25	2024-11-22 03:55	125.0	39	-3.17
35	2025-01-04 22:35	2025-01-25 00:55	20.0	41	-6.83
36	2025-02-25 00:20	2025-02-25 02:40	37.0	42	-9.33
37	2025-03-31 11:05	2025-04-01 04:25	147.0	-	-
38	2025-05-31 17:10	2025-06-01 09:15	666.0	44	-12.42
39	2025-08-25 13:55	2025-08-27 07:40	13.0	-	-
40	2025-11-10 11:25	2025-11-10 15:00	30.0	46	-6.42
41	2025-11-11 09:45	2025-11-12 02:15	1460.0	-	-
42	2025-11-14 09:20	2025-11-14 09:50	16.0	-	-

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